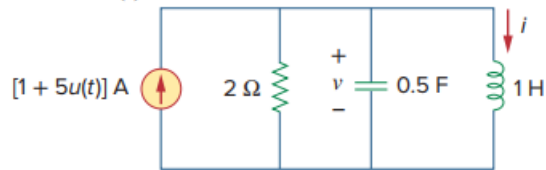


Problem

In the circuit of Fig. 8.92, find $v(t)$ and $i(t)$ for $t > 0$.



Step-by-step solution

Step 1 of 14

Refer to Figure 8.92 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is 1 A since the value $u(t)$ at $t < 0$ is 0 .

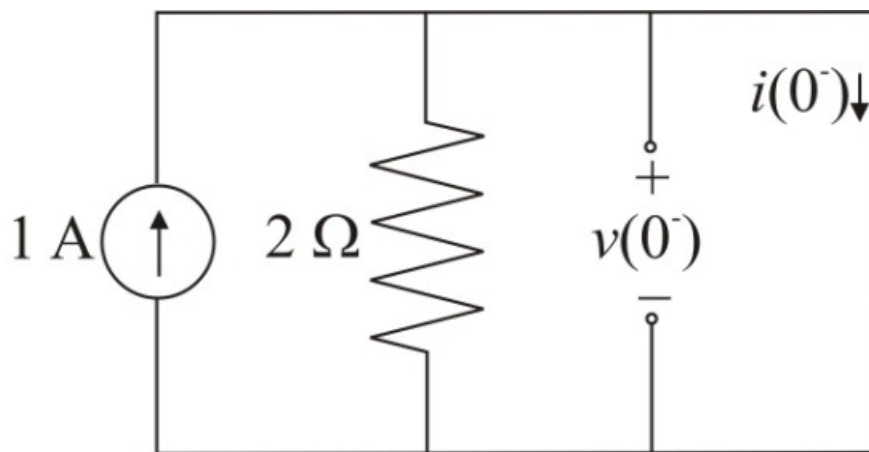


Figure 1

Notice that the resistor is shorted and no current passes through it. Hence, the value of the current $i(0^-)$ is 1 A and the value of the voltage source is 0 V .

Step 2 of 14

Draw the circuit diagram at $t > 0$.

The value of the current source is 6 A since $u(t)$ is 1 at $t > 0$.

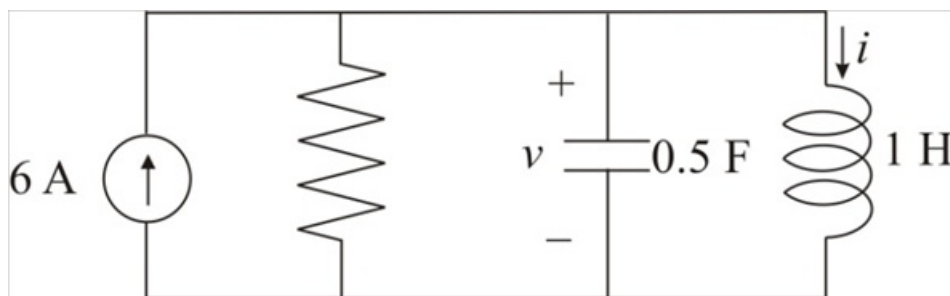


Figure 2

Step 3 of 14

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 0 V and the value of the current $i(0^+)$ is 1 A.

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 0$ V and $i(0) = 1$ A.

Step 4 of 14

Apply Kirchhoff's Current Law at the common node.

$$-6 + i + 0.5 \frac{dv}{dt} + \frac{v}{2} = 0$$

$$0.5 \frac{dv}{dt} = -\frac{v}{2} + 6 - i$$

$$\frac{dv}{dt} = -v + 12 - 2i$$

Calculate the value of $\frac{dv}{dt}$ for $t = 0$.

$$\frac{dv(0)}{dt} = -v(0) + 12 - 2i(0)$$

Substitute 0 V for $v(0)$ and 1 A for $i(0)$.

$$\begin{aligned} \frac{dv(0)}{dt} &= -0 + 12 - 2(1) \\ &= 12 - 2 \\ &= 10 \text{ V/s} \end{aligned}$$

Therefore, the value of $\frac{dv(0)}{dt}$ is 10 V/s.

Step 5 of 14

Deactivate the sources to estimate the transient response of the circuit.

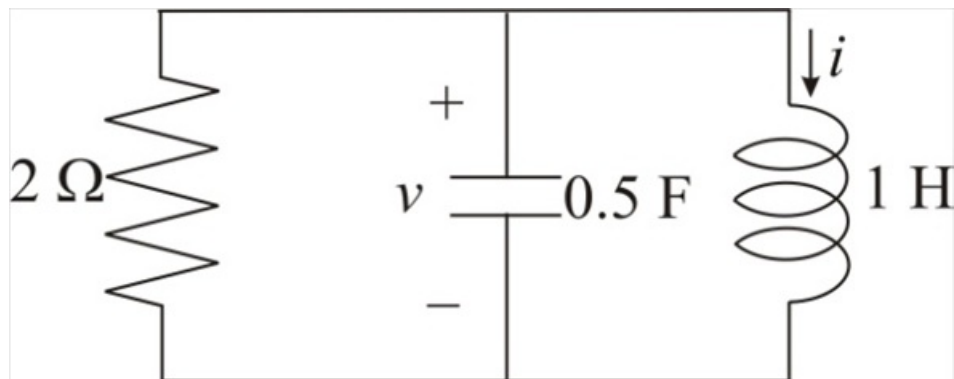


Figure 3

Step 6 of 14

Apply Kirchhoff's Current Law at the common node.

$$i + 0.5 \frac{dv}{dt} + \frac{v}{2} = 0$$

Substitute $\frac{1}{L} \int v dt$ for i .

$$\frac{1}{L} \int v dt + 0.5 \frac{dv}{dt} + \frac{v}{2} = 0$$

Substitute 1 H for L .

$$\frac{1}{1} \int v dt + 0.5 \frac{dv}{dt} + \frac{v}{2} = 0$$

$$\int v dt + 0.5 \frac{dv}{dt} + \frac{v}{2} = 0$$

Apply derivative with respect to t .

$$v + 0.5 \frac{d^2v}{dt^2} + \frac{1}{2} \frac{dv}{dt} = 0$$

$$\frac{d^2v}{dt^2} + \frac{dv}{dt} + 2v = 0$$

Treat $\frac{d}{dt}$ as s .

$$s^2v + sv + 2v = 0$$

$$(s^2 + s + 2)v = 0$$

Step 7 of 14

Consider the characteristic equation of the circuit.

$$s^2 + s + 2 = 0$$

$$(s + 0.5 + j1.323)(s + 0.5 - j1.323) = 0$$

The roots of the characteristic equation are $s = -0.5 \pm j1.323$.

The circuit exhibits under-damped response since the roots of the characteristic equation are complex.

Compare the roots to $s = -\alpha \pm j\omega_d$.

The values of α and ω_d are 0.5 Np/s and 1.323 rad/s respectively.

Step 8 of 14

Draw the circuit for $t = \infty$.

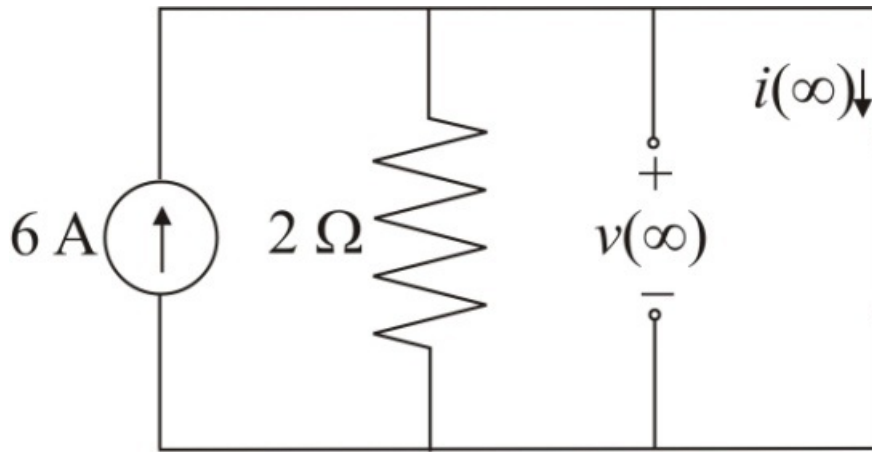


Figure 4

The value of the current $i(\infty)$ is 6 A and the value of the voltage $v(\infty)$ 0 V.

Step 9 of 14

Write the general solution for the voltage $v(t)$ for under-damped condition.

$$v(t) = v(\infty) + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

Substitute 0 V for $v(\infty)$, 1.3229 rad/s for ω_d and 0.5 Np/s for α .

$$v(t) = 0 + (A_1 \cos 1.3229t + A_2 \sin 1.3229t) e^{-0.5t} \quad v(t) = (A_1 \cos 1.3229t + A_2 \sin 1.3229t) e^{-0.5t}$$

..... (1)

Step 10 of 14

Recall equation-1.

$$v(t) = (A_1 \cos 1.3229t + A_2 \sin 1.3229t) e^{-0.5t}$$

Substitute 0 for t .

$$\begin{aligned} v(0) &= (A_1 \cos 1.3229(0) + A_2 \sin 1.3229(0)) e^{-0.5(0)} \\ &= (A_1 \cos(0) + A_2 \sin(0)) e^{(0)} \\ &= (A_1(1) + A_2(0))(1) \\ &= A_1 + 0 \end{aligned}$$

Substitute 0 V for $v(0)$.

$$A_1 = 0$$

Therefore, the value of the constant A_1 is 0.

Step 11 of 14

Recall equation-1.

$$v(t) = (A_1 \cos 1.3229t + A_2 \sin 1.3229t)e^{-0.5t}$$

Apply derivative with respect to t .

$$\frac{dv(t)}{dt} = \left\{ (A_1 \cos 1.3229t + A_2 \sin 1.3229t)(-0.5)e^{-0.5t} + e^{-0.5t}(-1.3229A_1 \sin 1.3229t + 1.3229A_2 \cos 1.3229t) \right\}$$

Substitute 0 for t .

$$\begin{aligned} \frac{dv(0)}{dt} &= \left\{ (A_1 \cos 1.3229(0) + A_2 \sin 1.3229(0))(-0.5)e^{-0.5(0)} + e^{-0.5(0)}(-1.3229A_1 \sin 1.3229(0) + 1.3229A_2 \cos 1.3229(0)) \right\} \\ &= (A_1 \cos(0) + A_2 \sin(0))(-0.5)e^{(0)} + e^{(0)}(-1.3229A_1 \sin(0) + 1.3229A_2 \cos(0)) \\ &= (A_1(1) + A_2(0))(-0.5)(1) + (1)(-1.3229A_1(0) + 1.3229A_2(1)) \\ &= -0.5A_1 + 1.3229A_2 \end{aligned}$$

Substitute 10 V/s for $\frac{dv(0)}{dt}$ and 0 for A_1 .

$$10 = (-0.5)(0) + 1.3229A_2$$

$$10 = 1.3229A_2$$

$$A_2 = \frac{10}{1.3229}$$

$$A_2 = 7.558$$

Therefore, the value of the constant A_2 is 7.559.

Step 12 of 14

Recall equation (1).

$$v(t) = (A_1 \cos 1.3229t + A_2 \sin 1.3229t)e^{-0.5t}$$

Substitute 7.559 for A_2 and 0 for A_1 .

$$\begin{aligned} v(t) &= ((0) \cos 1.3229t + 7.559 \sin 1.3229t)e^{-0.5t} \\ &= (0 + 7.559 \sin 1.3229t)e^{-0.5t} \\ &= 7.559e^{-0.5t} \sin 1.3229t \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $7.559e^{-0.5t} \sin 1.3229t \text{ V}$.

Step 13 of 14

Consider the expression for the current $i(t)$.

$$i(t) = \frac{1}{L} \int v(t) dt$$

Substitute $7.559e^{-0.5t} \sin 1.3229t$ V for $v(t)$ and 1 H for L .

$$\begin{aligned} i(t) &= \frac{1}{1} \int 7.559e^{-0.5t} \sin 1.3229t dt \\ &= 7.559 \int e^{-0.5t} \sin 1.3229t dt \end{aligned}$$

Apply the formula $\int e^{at} \sin bt dt = \frac{e^{at}}{a^2 + b^2} (a \sin bt - b \cos bt) + C$ for $\int e^{-0.5t} \sin 1.3229t dt$.

$$\begin{aligned} i(t) &= \frac{(7.559)e^{-0.5t}}{(-0.5)^2 + (1.3229)^2} (-0.5 \sin 1.3229t - 1.3229 \cos 1.3229t) + C \\ &= \frac{(7.559)e^{-0.5t}}{0.25 + 1.75} (-0.5 \sin 1.3229t - 1.323 \cos 1.3229t) + C \\ &= \frac{(7.559)e^{-0.5t}}{2} (-0.5 \sin 1.3229t - 1.3229 \cos 1.3229t) + C \end{aligned}$$

$$i(t) = (3.779)e^{-0.5t} (-0.5 \sin 1.3229t - 1.3229 \cos 1.3229t) + C \dots\dots (2)$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= (3.779)e^{-0.5(0)} (-0.5 \sin 1.3229(0) - 1.3229 \cos 1.3229(0)) + C \\ &= (3.779)e^{(0)} (-0.5 \sin(0) - 1.3229 \cos(0)) + C \\ &= (3.779)(1) (-0.5(0) - 1.3229(1)) + C \\ &= (3.779)(-1.3229) + C \end{aligned}$$

Substitute 1 A for $i(0)$.

$$1 = -5 + C$$

$$C = 1 + 5$$

$$C = 6$$

Therefore, the value of C is 6.

Step 14 of 14

Recall equation (2).

$$i(t) = (3.779)e^{-0.5t} (-0.5 \sin 1.3229t - 1.3229 \cos 1.3229t) + C$$

Substitute 6 for C .

$$\begin{aligned} i(t) &= (3.779)e^{-0.5t} (-0.5 \sin 1.3229t - 1.3229 \cos 1.3229t) + 6 \\ &= 6 - (5 \cos 1.3229t + 1.8895 \sin 1.3229t)e^{-0.5t} \text{ A} \end{aligned}$$

Therefore, the value of the current $i(t)$ is

$$\boxed{6 - (5 \cos 1.3229t + 1.8895 \sin 1.3229t)e^{-0.5t} \text{ A}}.$$